Marika Nishi Robotics Engineering Intern

mari3511n@gmail.com (445) 256-0089 Philadelphia, PA linkedin.com/in/marika-nishi/ github.com/marika340

EDUCATION

University of Pennsylvania, School of Engineering & Applied Science | Philadelphia, PA, United States *May 2027*
Candidate for Master of Science in Engineering, *Major: Robotics, Computer Science* | *Concentrations: Autonomous Driving* GPA: 3.78/4.0
University of Tokyo, Faculty of Engineering | Bunkyo, Tokyo, Japan *March 2023*
Bachelor of Engineering, *Major: Mechanical Engineering* GPA: 3.64/4.0

SKILLS & RELEVANT COURSEWORK

Software Skills: C++, Python, Linux, ROS, Pytorch, ML-based Planning, Machine Learning, Simulator (Gazebo, MuJoCo), Control, OpenCV, SLAM, Sensor Fusion, Object Detection, Segmentation, Matlab, Autonomous driving, Git, Docker
Hardware Skills: CAD, mechanical design, test and measurement, LiDAR, RGBD camera, GPS, IMU, Additive Manufacturing
Other Skills: Leadership, Research, Design and development, Report writing, Teamwork, Troubleshooting
Relevant Coursework: Mechatronics; Automotive Engineering; F1tenth Autonomous Racing Cars; Machine Perception; Computer Vision; Machine Learning; Advanced Robotics; Dynamics and Control; Deep Generative Models
Publication: Watanabe, K., **Nishi, M.**, & Ito, T. (2025). Placement design of various roadside sensors for stochastic position prediction of traffic participants on community roads. *International Journal of Intelligent Transportation Systems Research.*

EMPLOYMENT EXPERIENCE

Vijay Kumar Lab | *Research Intern,* Philadelphia, PA, United States *May 2025 – Present*

- Implement driving simulators that visualized data of event camera on F1tenth car using Gazebo, CARLA, and ROS
- Collect datasets on driving simulator and train vision transformer model that computed appropriate motion control from event camera images; improve data generation process efficiency by removing redundancy, cutting time by 70 %

ModLab | *Summer Research Intern,* Philadelphia, PA, United States *May 2025 – August 2025*

- Built ROS2 computer vision system that detected surgical needle posture using color identification and April Tag

BOSCH | *Software Intern*, Tsuzuki, Kanagawa, Japan *August 2023 – September 2023*

- Designed and built map to enhance autonomous driving by visualizing key information, such as routes and vehicle state using MATLAB; helped company's engineers understand signals sent from cars visually
- Improved BOSCH's software that processed location CAN signals from cars by fixing CAN signal decoding; refined accuracy of system outputs
- Collaborated with colleagues from 10 countries and different majors; made outcomes reflecting diverse opinions

DMG MORI | *Industrial Practices Intern, Additive Manufacturing R & D*, Iga, Mie, Japan *August 2021 – September 2021*

- Analyzed performance tests of additive manufacturing machines by operating the machines and inspecting products
- Investigated additive manufacturing methods by examining machine structures; proposed solutions to heat distortion

ROBOT AUTOMATION PROJECTS

F1tenth Autonomous Racing Cars | *Perception, Planning, Control, Autonomous Driving, Python, C++, ROS2* *January 2025 – May 2025*

- Developed autonomous racing car that was 1/10 scale of F1 racing cars; designed and implemented motion planning and vehicle control systems using Python, C++, ROS, SLAM localization; won second place for fastest lap in race

Research of Traffic Collision Prediction System | *Prototyping, Extended Kalman Filter , C++, ROS* *April 2022 – March 2023*

- Designed and built real-time collision prediction system among pedestrian, cyclist, and intelligent wheelchair
- Wrote C++ ROS codes that detected pedestrians and cyclists based on LiDAR data
- Conducted sensor fusion using Extended Kalman Filter; estimated pedestrians and cyclists' positions
- Operated RTK-GPS, low-accuracy GPS, LiDAR, and IMU to collect and integrate data into system development